Practical impact: who is affected, what is measured, and what is only intended

University GPU cluster, 100 GPU-hours, six teams. Official SDG icons; targets verified at sdgs.un.org/goals.

Demonstrated in simulation

Intended: no adoption yet



Target 9.5

Enhance scientific research and technological capability

Mechanism: GPU-hours go to the group with the most project value net of carbon, not to the earliest arrivals.

Indicator: carbon-adjusted research value served

Baseline	Result
FCFS mean 16.96	VCG 19.2 (+13%)

Demonstrated: computed on one synthetic instance



Target 9.4

Resource-use efficiency and clean technology in infrastructure

Mechanism: carbon enters each team's score, so large jobs move to the cleaner GPU pool when one exists.

Indicator: kg CO_2e per unit of allocated value

Baseline	Result
FCFS 0.81–0.84	VCG 0.72 (two pools 0.55)

Demonstrated: computed on one synthetic instance



Target 13.3

Institutional capacity on climate change mitigation

Mechanism: the institution publishes λ, issues a fixed credit budget, and keeps a re-runnable allocation log.

Indicator: rounds with λ, credit supply, and log published in advance

Baseline	Status
none today	planned pilot

Intended: requires an institution to adopt the rule

Affected stakeholders

Research teams
receive compute by value; pay a carbon charge and a VCG price in credits

Cluster administrators
must justify each allocation; can re-run it from public inputs

The institution
sets λ, issues and retires credits, owns the audit log

The public
bears the carbon cost whoever is served

Risk	Safeguard
A heavy λ penalizes compute-heavy research (9.4 works against 9.5): above λ = 1.25 only two teams are served.	Publish λ before each round and cap it below 1.25, the bound found by the sweep.
The same low-value teams are excluded every round (VCG serves 4 teams; FCFS 4.93 on average).	History-based priority credits for repeatedly excluded teams (planned; Roadmap, CS study).
Adding credits mid-term removes their cost, and with it the reason to report truthfully.	Fixed credit supply per round (C2); no top-ups; spent credits are retired.
True values cannot be verified, and a misreport that wins a seat can displace an honest team.	Public, re-runnable audit log; a pilot comparing reported value with ex-post outcomes.